

Nixtamal Tortillas

Stone-ground from whole kernel · El Milagro style · cal-nixtamalized corn

YIELD	TIME	CAL RATIO	MASA MOISTURE
40–45 tortillas	30 min + overnight	1% dry corn wt	55–60%

INGREDIENTS

Cal = calcium hydroxide / food-grade pickling lime

Yellow dent corn Tuxpeño or similar, dry	680 g
Water for cooking, ~3:1 by volume	2 L
Calcium hydroxide cal / pickling lime, 1% of corn	7 g
Water for hydration, during grind	as needed
Salt optional, in the final knead	pinch

EQUIPMENT

A stone wet-grinder is what makes the masa silky

Wet grinder	Premier Small Wonder (1.5 L) — stone grinding for silky masa
Pot + strainer	Large, non-reactive (stainless)
Scale	Digital — essential for the ratios
Press	Tortilla press + parchment circles (or plastic sheets)
Comal	Comal / cast iron, preheated to 200–230°C (400–450°F)

STAGE 1

NIXTAMALIZATION

Cook + overnight steep — the chemistry happens here

- 1 Rinse:** rinse corn under cold water; remove debris and broken kernels.
- 2 Dissolve:** stir **7 g cal** into **2 L water** — it goes milky/cloudy. Fully dissolved before the corn goes in.
- 3 Cook:** add corn, bring to a boil (~10 min), then gentle simmer **15–25 min**. Done when the pericarp (skin) **slips off easily** when rubbed and the kernel is soft.
- 4 Steep:** off the heat, covered, **8–12 hr** (overnight ideal) — the cal keeps penetrating.
- 5 Wash:** drain, then rinse **3–5x**, rubbing to shed the loosened skins, until the water runs mostly clear. **Keep it damp — do not dry**. Yields ~1,500 g wet nixtamal.

Grind, Press & Cook

STAGE 2

GRIND & HYDRATE

~1,400 g masa — Play-Doh texture is the goal

- 1 Load:** the grinder holds ~750 g wet nixtamal — load the first half, **no extra water yet** (inherent moisture is ~45–50%).
- 2 Grind:** run **10–15 min**, scraping the sides every 2–3 min. A coarse paste forms from the kernel moisture alone.
- 3 Hydrate:** add warm water incrementally — **15–30 ml per half, 5 ml at a time**, kneading between — to **55–60% moisture**.
- 4 Test:** smush a ball flat — **smooth edges, no cracks**; earlobe-soft; clean release from your hands; a pea-sized ball floats.
- 5 Combine:** scrape out, repeat with the second half, combine, knead **2–3 min**. Add a pinch of salt if using.
- 6 Rest:** cover with a damp towel, rest **10–20 min** — hydration equalizes and flavor develops.

STAGE 3

PRESS & COOK

Three flips — the third is the puff

- 1 Portion:** weigh **32 g balls** (~40–45 total), roll smooth, keep covered.
- 2 Press:** ball between parchment/plastic — press once, rotate 180°, press again for even **~1.5 mm** thickness, no thin spots.
- 3 First side:** onto the preheated dry comal; cook until the edges dry and lift slightly (~15 sec).
- 4 Flip 1:** cook until the color changes and light spots appear across the surface (~40 sec).
- 5 Flip 2 (puff):** flip back to the first side — it should **puff**; a gentle press on the edges with a towel helps (10–15 sec).
- 6 Stack:** transfer to a towel-lined container, cover immediately — the steam keeps them pliable.

QUALITY TARGETS

Four of the five are really one question — is the masa at 55–60%?

TEST	WHAT YOU WANT	IF NOT
Smush test	Edges smooth, no cracks	Add water 5 ml at a time
Earlobe test	Same softness as an earlobe	Too dry: add water. Too wet: add masa.
Float test	Pea-sized ball floats	Under-hydrated — knead in more water
Crumple test	Unfurls without cracking	Overcooked or under-hydrated
Puff	Full inflation on the third flip	Comal temp, thickness, or hydration is off

STORAGE

Reheat on the comal, never cold

METHOD	DURATION	NOTES
Towel-lined container, room temp	4 hr	Best fresh
Paper bag, refrigerated	5–7 days	Reheat on the comal 10 sec/side
Freezer, stacked + separated	1–2 months	Thaw, then comal — or damp towel + microwave